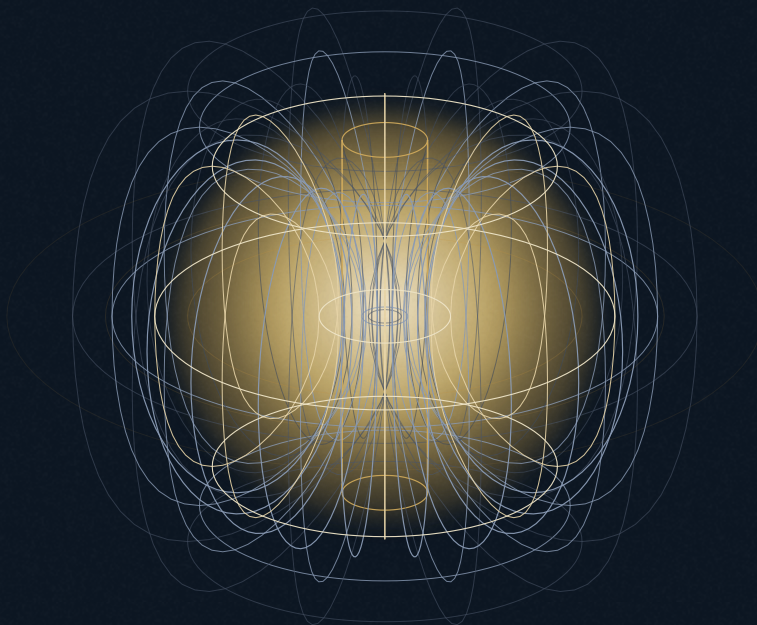




KRONOS FUSION ENERGY



PAPER 1.1 OF THE 2026 KRONOS SET

Hyperion

The Tritium Foundry

A compact negative-triangularity spherical-tokamak breeder, sized to a national strategic-materials requirement rather than to a power rating.

Spherical tokamak · D-T · Q 3.0763 · 85.04 MW_{fus} · ~2.0 kg-T/yr/unit

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

Hyperion — The Tritium Foundry

Hyperion is a compact negative-triangularity spherical-tokamak breeder that earns on fusion *gain* alone — producing tritium, decay helium-3, and 14 MeV neutrons — so it delivers value without waiting on the net-electricity milestone the rest of the field is chasing. At the frozen reference point it reaches $Q = 3.0763$ at 85.04 MW and 9.66 MA, with negative triangularity cutting ion-scale transport by 40–45% into a quiet subcritical core. The tritium output is a confirmed ladder to net TBR 1.42 (~ 2.0 kg-T yr⁻¹ per unit); a single advanced unit is an exporter, a national 4 kg-T yr⁻¹ supply a fleet property. The centrepost is a quantified consumable and the disruption response a stated hard requirement.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132235](https://doi.org/10.5281/zenodo.22132235) (v2 [10.5281/zenodo.21795620](https://doi.org/10.5281/zenodo.21795620)) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords: spherical tokamak tritium breeding negative triangularity strategic isotopes consumable centrepost

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

Why a breeder, and why it can be small

United States tritium supply is structurally fragile: weapons-stockpile tritium is produced through a *single* commercial reactor, and the federal replacement requirement — tritium decays at **5.5% per year** — has been met by increasing irradiation loading rather than by adding a second source. Helium-3 is rationed. A domestic, non-reactor capability to produce both isotopes is of strategic interest independently of whether it is inexpensive.

The obstacle to a *fusion* isotope source has never been the nuclear physics, which is textbook, but the perceived need for a power-plant-scale machine. Hyperion removes that obstacle with one observation: **a breeder needs fusion gain — reactions — not net electricity**. Demanding net electricity is what forced every prior breeder concept into a large plasma, a large blanket, and a dangerous standing tritium inventory. Freed from that requirement, the machine is sized to a national supply requirement of order kilograms per year — and the size that results is small.

THE THESIS IN ONE LINE

Size the breeder to a named national requirement, not to a power rating, and the two obstacles that defeated every prior fusion-breeder proposal — machine scale and tritium inventory — both fall below their binding thresholds. The physics never forbade a small breeder; it forbade a large one.

The fusion helium-3/tritium breeder has been proposed before and rejected in print — Wittenberg on a licensing/inventory basis, Greenspan & Miley on the merchant-breeder economics — but every prior study sizes the breeder to a *fusion fleet's* support ratio, where the inventory objection bites. Hyperion sizes to a named non-fusion requirement, at which the objection inverts.

The design point and the power balance

The zero-dimensional power balance fixes the gain. At the frozen operating point (config **22021**, gate BR-L1-A1) the compact spherical tokamak reaches a plasma gain of $Q = 3.0763$ at a fusion power of $P_{\text{fus}} = 85.04$ MW and a plasma current of $I_p = 9.66$ MA, at negative triangularity $\delta = -0.3$. This is a driven, sub-ignition target — it does not need to close the electrical loop, only to run reactions at a useful rate.

COMPUTED · FROZEN

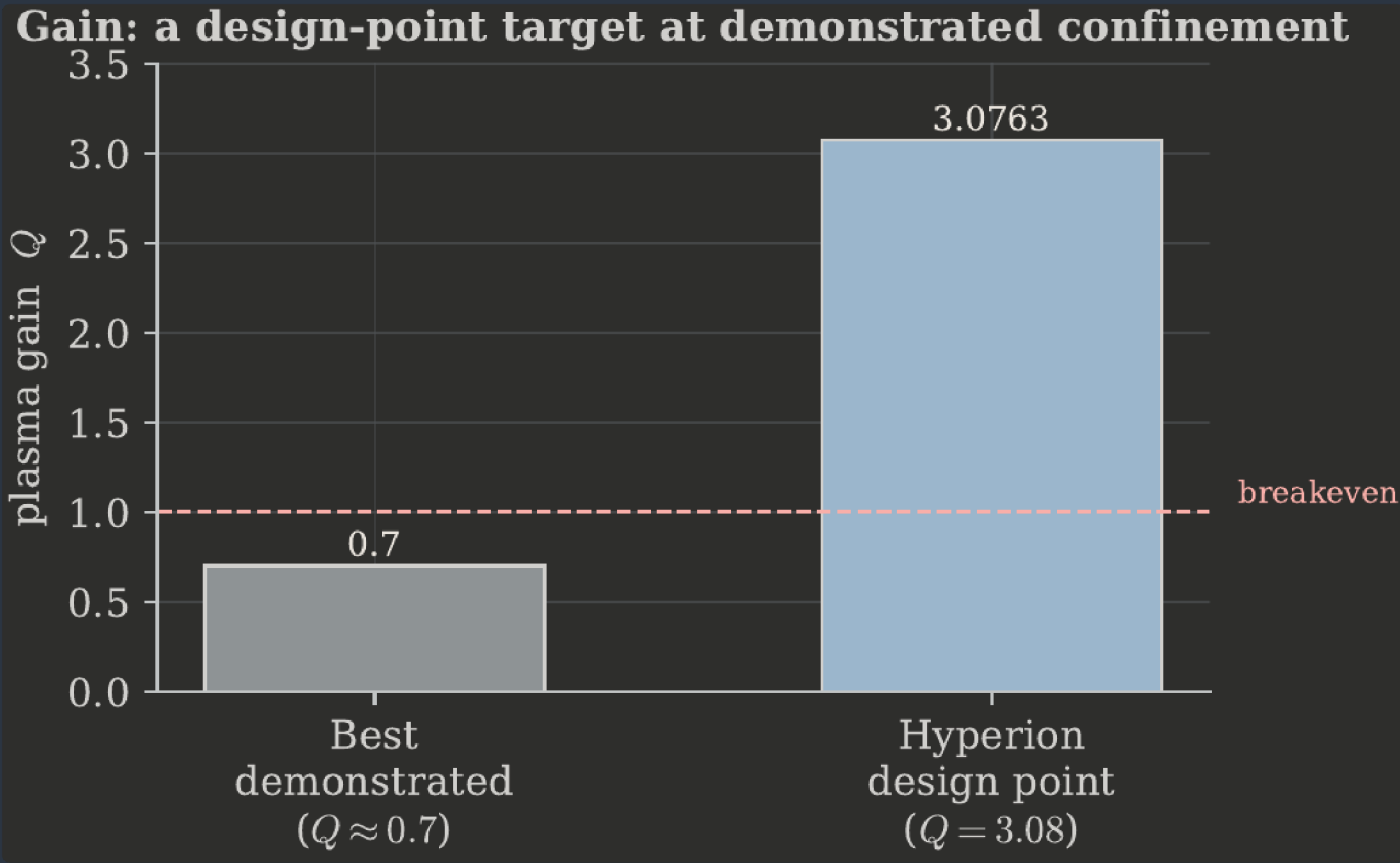


FIGURE 1. The power balance at the frozen operating point: fusion gain $Q = 3.0763$ at 85.04 MW, a driven sub-ignition breeder, not a power reactor.

RECOMPUTED

§ 3

Negative triangularity — the confinement lever

Negative triangularity is the design's confinement lever, established at both turbulence scales with nonlinear gyrokinetics. At the operating gradient it cuts the combined ITG+TEM ion transport by 40–45% relative to a positive-triangularity twin (gate BR-L2-A1c), and suppresses electron-scale ETG transport by 79–80% — leaving the operating point quiet and subcritical. The direction is corroborated against the TCV and DIII-D negative-triangularity campaigns; the magnitude is a design-target caveat carried openly.

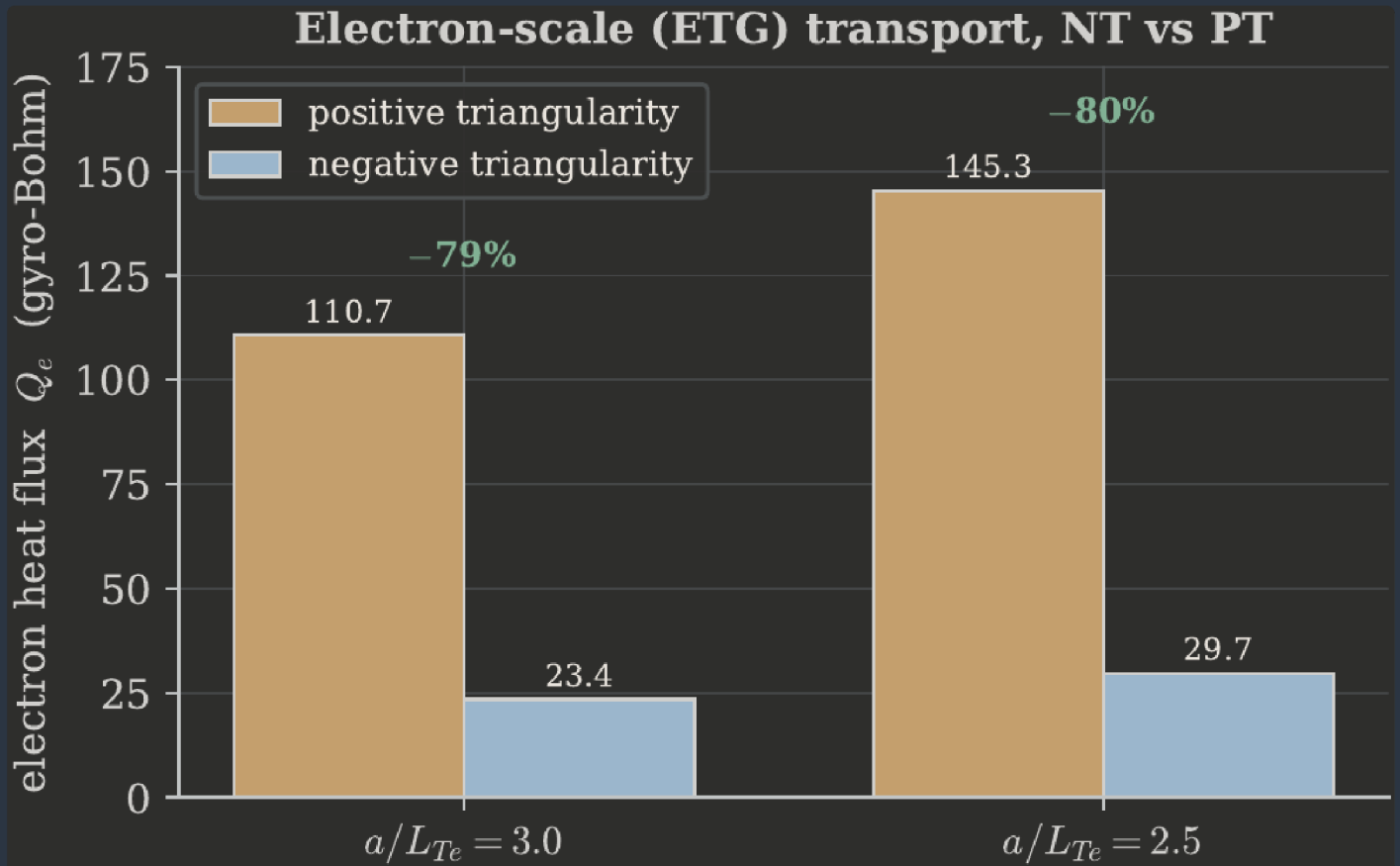


FIGURE 2. Negative triangularity suppresses turbulent transport at both the ion and electron scale relative to a positive-triangularity twin. NONLINEAR CGYRO

The tritium breeding ladder

The breeder’s product is tritium, and its credibility rests on an honest mass balance. Each deuterium–tritium fusion consumes one triton, so the per-unit burn rate follows from the fusion power: $\Lambda_T = 4.76 \text{ kg-T fpy}^{-1}$. The net surplus is a confirmed *ladder*: three-dimensional neutronics on ENDF/B-VIII.0 raise it from 0.29 kg-T yr^{−1} (literal recipe) through 0.9 (solid-beryllium blanket) to ~2.0 kg-T yr^{−1} per unit at the advanced blanket — net TBR 1.42, the confirmed per-unit ceiling across twelve configurations.

PER UNIT VERSUS FLEET

A single advanced unit is a tritium **exporter** at the ~2.0 kg-T yr^{−1} scale. A national-scale 4 kg-T yr^{−1} supply is a *fleet* property — two advanced units, or four-to-five solid-beryllium units, with helium-3 aggregating to 3.6-4.5 kg yr^{−1} — never a single-unit claim, because the spherical-tokamak centrepost caps the per-unit ceiling.

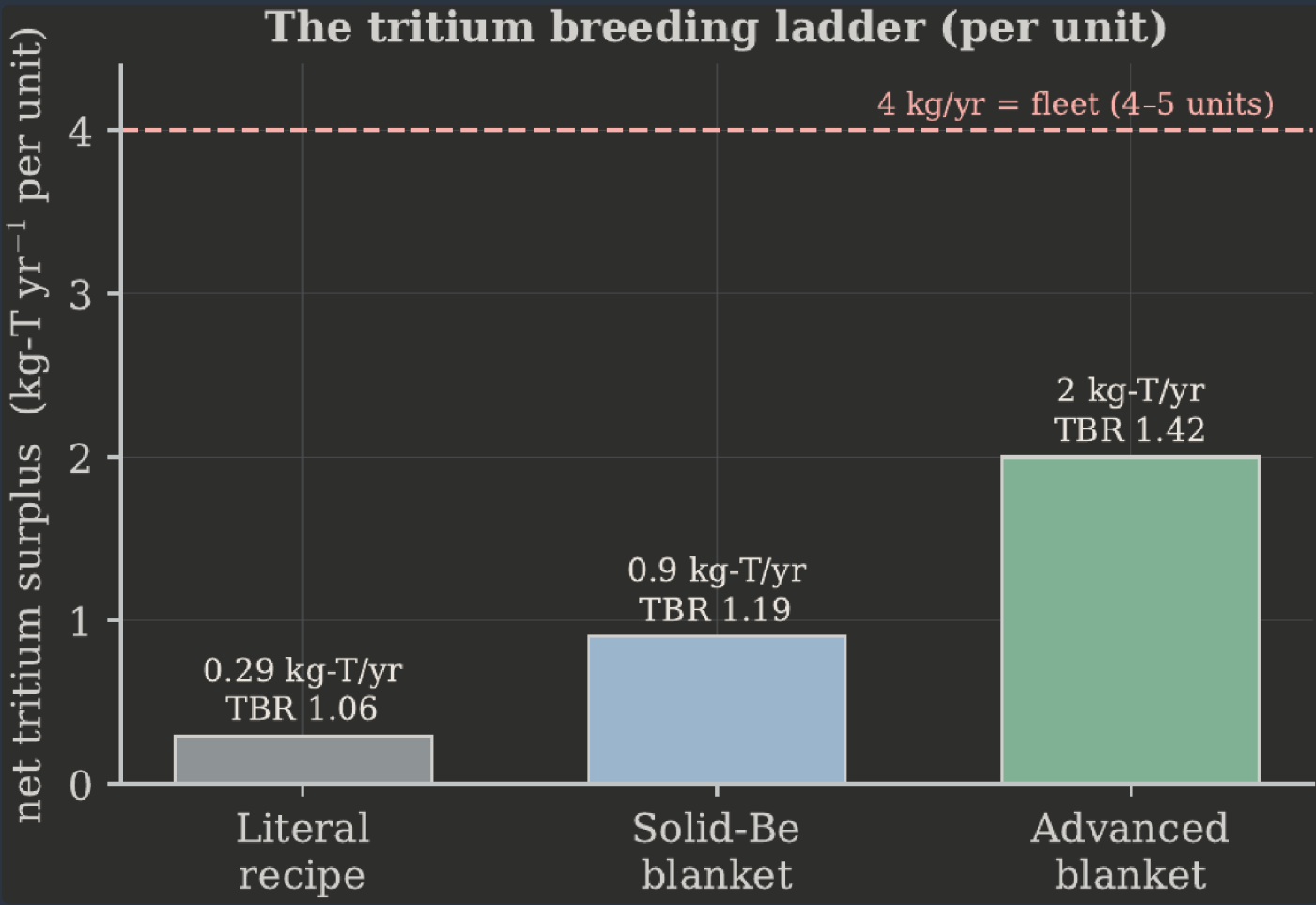


FIGURE 3. The per-unit tritium breeding ladder: net surplus at the literal, solid-beryllium, and advanced blankets. [OPENMC / ENDF-B-VIII.0](#)

The centrepost, as a quantified consumable

The inboard centrepost is the defining vulnerability of a normal-conductor spherical tokamak, and the design does not argue it away: it is a consumable, scheduled-replacement cartridge with a best-case life of **0.58 full-power-years**, tunable against a tungsten-carbide shield that trades directly with breeding coverage. The materials-lifetime problem is converted into a maintenance-interval problem, which is the tractable one.

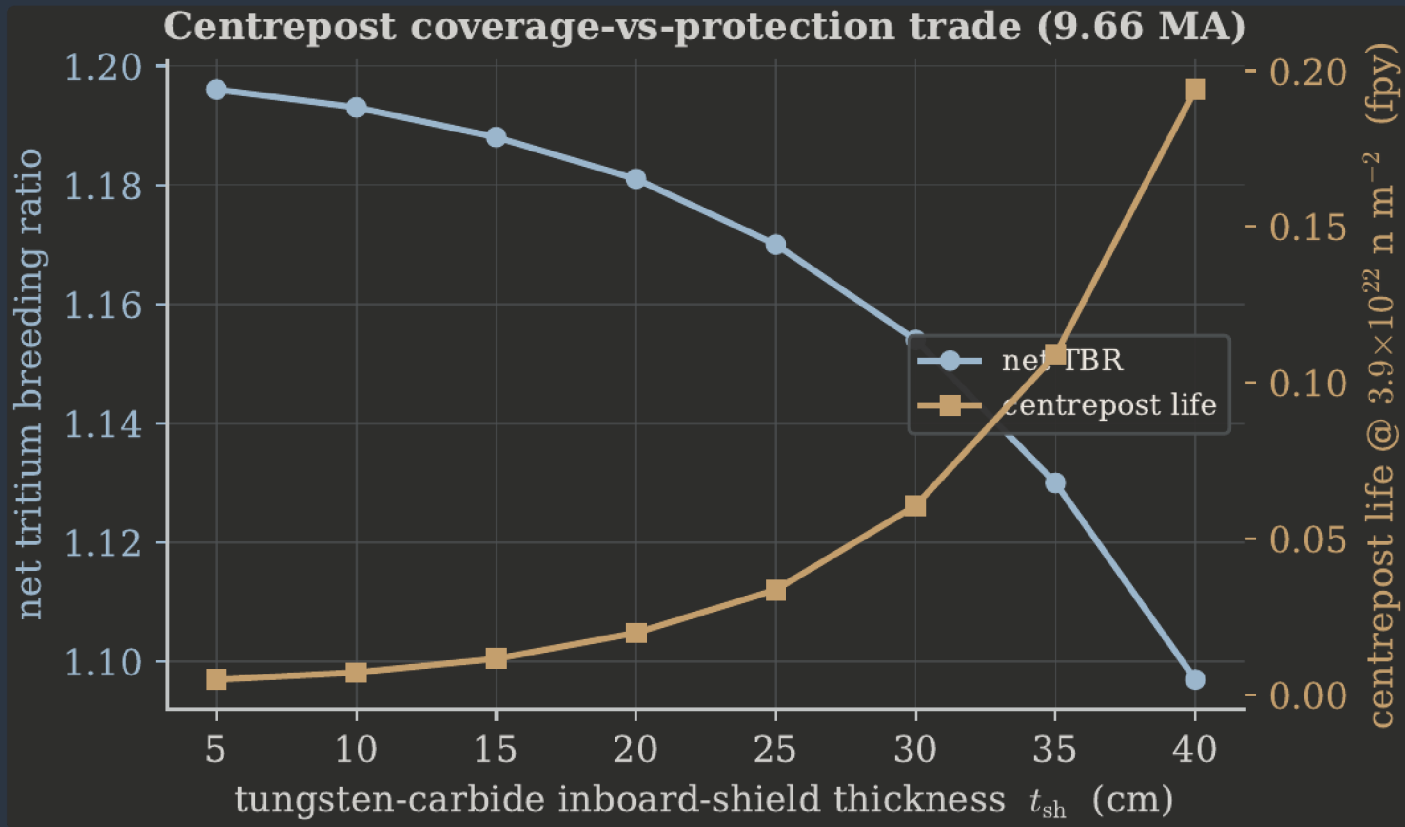


FIGURE 4. Centrepost damage and consumable-cartridge life against shield thickness; the shield trades against breeding coverage. RECOMPUTED

Disruptions, answered directly

The disruption response is answered rather than assumed negligible, and it is demanding. On a NIMROD-verified equilibrium the current-quench time is 7-19, the runaway-electron avalanche is *severe* ($E/E_c \approx \text{SEVERE}$), and the halo current reaches 35 of the plasma current. The honest conclusion is a hard design requirement: shattered-pellet-class disruption mitigation, triggered inside the current-quench window. The risk is stated as a requirement, not waved away. **DESIGN REQUIREMENT**

Net electricity is a category error for this machine and is excluded. Hyperion succeeds by gain, breeding ratio, helium-3, and neutrons — the materials a fusion economy needs — and the burner, its companion, carries the power story.

A P P A R A T U S

Portfolio, People & Record

*The intellectual-property estate, the people who built it, the limitations
that gate it, and the sources it rests on.*

1

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

Explore it live — turn the interactive [3-D model](#), run the deposited solvers in the [physics validator](#), and watch the [companion film series](#).



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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry (<i>this paper</i>) · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

Explore the whole programme live: the interactive [3-D model](#), the in-browser [physics validator](#), and the [companion film series](#).



KRONOS FUSION ENERGY

Hyperion

Explore the science — live, in your browser.

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